

Sequence Modelling for Translation, Denoising, and Generation: A Study of RNN, DAE, and VAE Architectures

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Abstract. This report presents implementations and evaluations of three generative neural network architectures for distinct tasks: (1) a vanilla Recurrent Neural Network (RNN) Encoder-Decoder for English-to-Urdu machine translation, (2) a Convolutional Denoising Autoencoder (DAE) for image restoration on CIFAR-10, and (3) a Variational Autoencoder (VAE) for generative modelling on Fashion-MNIST. For translation, the model was trained on 2,019 short sentence pairs extracted from a 9,093-pair corpus and evaluated using BLEU score and qualitative error analysis. The DAE was evaluated under Gaussian and salt-and-pepper noise at multiple noise levels using PSNR and SSIM metrics, achieving a best PSNR of 23.46 dB. The VAE was trained across multiple latent dimensionalities and assessed via reconstruction fidelity and the quality of novel samples drawn from the learned prior. Key findings confirm the well-documented limitations of vanilla RNNs for translation and demonstrate the effectiveness of convolutional autoencoders and VAEs for vision tasks.

Keywords: Machine Translation · Denoising Autoencoder · Variational Autoencoder · RNN · BLEU · PSNR

1 Introduction

Sequence-to-sequence models, autoencoders, and variational generative models form foundational pillars of modern deep learning. This assignment explores all three paradigms under controlled conditions: a low-resource translation setting that stresses the limits of unattended RNNs, a corrupted-image reconstruction task that measures representational capacity, and a generative modelling problem that probes the quality of a learned latent space.

The three objectives are deliberately chosen to highlight both the strengths and failure modes of each architecture class, motivating more sophisticated designs (attention mechanisms, diffusion models) studied in later units.

2 Question 1: English-to-Urdu Machine Translation

2.1 Dataset and Preprocessing

The English-Urdu parallel corpus [1] contains 9,102 sentence pairs. After removing 9 duplicates and filtering sentences longer than 15 words, 2,019 usable pairs remained. These were split 80/10/10 into train (1,615), validation (201), and test (203) sets with no overlap confirmed programmatically. Vocabulary sizes after capping at 5,000 tokens per language were 2,438 (English) and 3,011 (Urdu). Special tokens `<pad>`, `<sos>`, `<eos>`, and `<unk>` were added to both vocabularies.

2.2 Model Architecture

A vanilla RNN Encoder-Decoder was implemented in PyTorch. The encoder maps a source sequence to a fixed-size hidden vector, which initialises the decoder. The decoder generates target tokens one step at a time using greedy or beam-search decoding at inference.

- **Encoder:** Embedding(2438, 256) $\rightarrow$ RNN(256, 512, layers=2)
- **Decoder:** Embedding(3011, 256) $\rightarrow$ RNN(256, 512, layers=2) $\rightarrow$ Linear(512, 3011)
- **Parameters:** 4,778,691
- **Weight init:** Orthogonal (RNN weights), zeros (biases)

2.3 Training Setup

Hyperparameters were selected via a grid search over four configurations (Table 1). The best configuration (**emb_dim=256**, **hid_dim=512**, **n_layers=2**, **lr=0.001**, **dropout=0.3**) was used for full training over 30 epochs with Adam optimiser, teacher forcing decaying linearly from 1.0 to 0.5, cross-entropy loss (padding tokens ignored), and best-checkpoint saving on validation loss.

Table 1. Hyperparameter grid search results (Q1).

emb_dim	hid_dim	layers	lr	dropout	Val Loss
128	256	1	0.003	0.1	7.8065
256	256	1	0.001	0.1	7.1212
256	512	2	0.001	0.3	6.8707
128	512	1	0.003	0.1	8.4002

2.4 Results and Evaluation

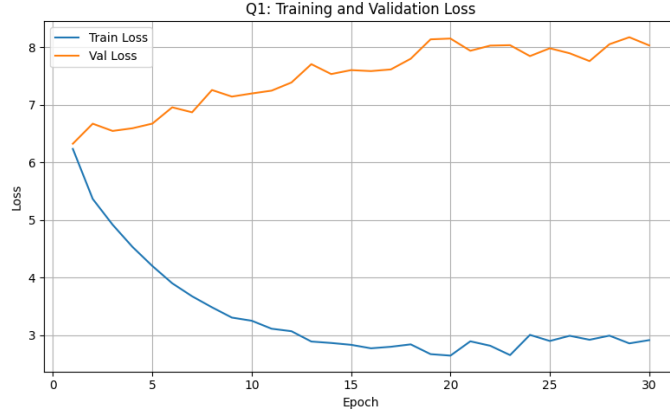


Fig. 1. Q1 training and validation loss over 30 epochs.

The training loss decreased from 6.23 to 2.92, while validation loss increased from 6.32 to 8.03, with best checkpoint at epoch 1 (val loss 6.32). Greedy BLEU was 0.0162 and beam-search ($k=3$) BLEU was 0.0180 on the test set (Table 2).

Table 2. Translation BLEU scores on test set (Q1).

Decoding Strategy BLEU	
Greedy	0.0162
Beam Search ($k = 3$)	0.0180

2.5 Error Analysis

Qualitative analysis of 30 test predictions revealed word_repetition in 30/30 cases — the model consistently produced high-frequency Urdu function words (, , ,) irrespective of input.

Root causes:

1. **Vanishing gradients:** Vanilla RNN gradients decay exponentially with sequence length; the decoder hidden state collapses to a near-constant attractor.
2. **Fixed context bottleneck:** A single vector must encode the entire source sentence; without attention, the decoder has no mechanism to align to relevant source positions.

3. **Severe overfitting:** With only 1,615 training pairs, the train/val loss gap widened monotonically (Fig. 1).
4. **Exposure bias:** Teacher-forced training and autoregressive inference operate under different token distributions.

Future improvements: Bahdanau attention, LSTM/GRU gating, subword tokenisation (BPE), Transformer architecture, and a larger parallel corpus (100k+ pairs) would each substantially improve translation quality.

3 Question 2: Image Denoising with Convolutional DAE

3.1 Dataset

CIFAR-10 [2] (60,000 RGB images, 32×32 pixels, 10 classes) was used. Images were normalised to $[0, 1]$. Two noise types were studied:

- **Gaussian noise:** $x_{\text{noisy}} = \text{clip}(x + \mathcal{N}(0, \sigma^2), 0, 1)$
- **Salt-and-pepper noise:** random pixels set to 0 or 1 with probability p

3.2 Model Architecture

A symmetric convolutional DAE with configurable bottleneck channels:

Encoder: Conv2d(3,32) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ MaxPool $\rightarrow$ Conv2d(32,64) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ MaxPool

Decoder: ConvTranspose2d(64,32) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ ConvTranspose2d(32,3) $\rightarrow$ Sigmoid

3.3 Training Setup

Both noise types were trained separately (15 epochs, Adam, lr=0.001, MSE loss). Both training and validation MSE were tracked per epoch. An experimental study varied noise factor across 0.1, 0.3, 0.5, 0.7 and bottleneck channel count across 16, 32, 64, 128.

3.4 Results

Table 3. DAE reconstruction quality on test set (Q2).

Noise Type	Noise Factor	PSNR (dB)	SSIM
Gaussian	0.3	21.74	0.944
Salt-Pepper	0.3	23.46	0.964

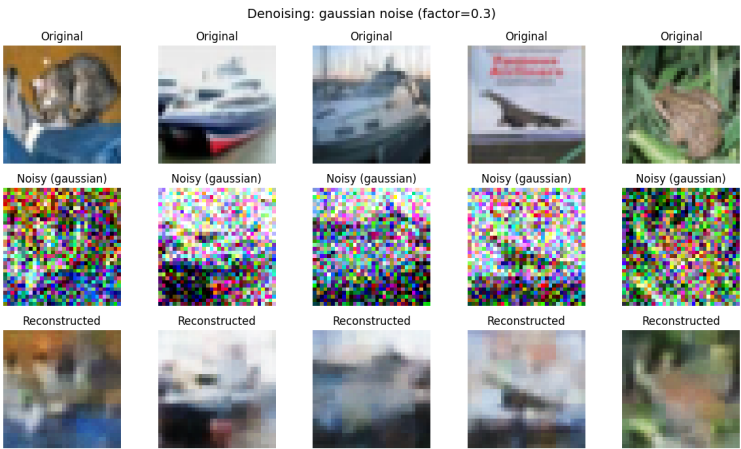


Fig. 2. Q2 Gaussian denoising: Original / Noisy / Reconstructed.



Fig. 3. Q2 Salt-and-pepper denoising: Original / Noisy / Reconstructed.

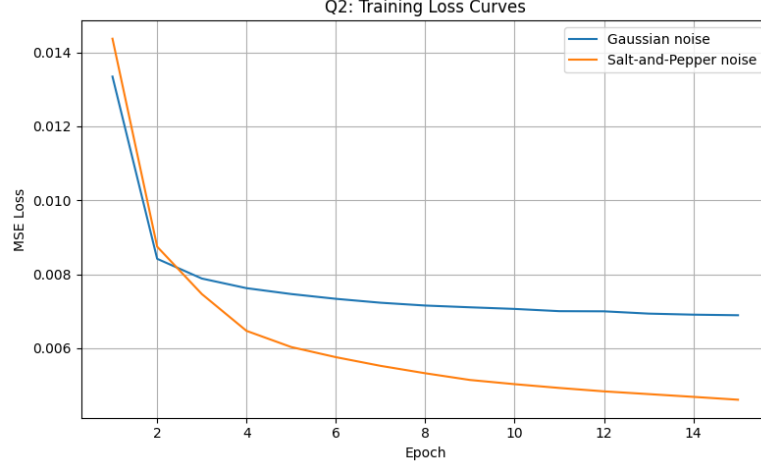


Fig. 4. Q2 training and validation MSE loss curves for both noise types.

The DAE effectively removed both noise types, recovering recognisable object structure from nearly fully corrupted inputs (middle row, Fig. 2). Salt-and-pepper noise achieved higher PSNR because its pixel-level corruption is more locally predictable. Output blurriness is characteristic of MSE-trained autoencoders, which optimise for average pixel error.

4 Question 3: Variational Autoencoder on Fashion-MNIST

4.1 Dataset

Fashion-MNIST [3] (70,000 grayscale images, 28×28 pixels, 10 clothing categories). Pixel values normalised to $[-1, 1]$.

4.2 Model Architecture

Encoder: $\text{FC}(784 \rightarrow 512) \rightarrow \text{ReLU} \rightarrow \text{FC}(512 \rightarrow 256) \rightarrow \text{ReLU} \rightarrow \text{FC}_\mu(256 \rightarrow z) + \text{FC}_\sigma(256 \rightarrow z)$

Reparameterisation: $z = \mu + \epsilon \cdot e^{0.5\sigma^2}$, $\epsilon \sim \mathcal{N}(0, I)$

Decoder: $\text{FC}(z \rightarrow 256) \rightarrow \text{ReLU} \rightarrow \text{FC}(256 \rightarrow 512) \rightarrow \text{ReLU} \rightarrow \text{FC}(512 \rightarrow 784) \rightarrow \text{Tanh}$

4.3 Loss Function

$$\mathcal{L} = \underbrace{E_{q(z|x)}[\log p(x|z)]}_{\text{Reconstruction (BCE)}} - \underbrace{D_{KL}(q(z|x)||p(z))}_{\text{Regularisation}} \quad (1)$$

The KL divergence term has the closed-form solution: $-\frac{1}{2} \sum (1 + \log \sigma^2 - \mu^2 - \sigma^2)$.

4.4 Training Setup

Trained 20 epochs, Adam (lr=0.001), batch size 128. An experimental study compared latent dimensions $z \in \{2, 10, 20, 50\}$.

4.5 Results

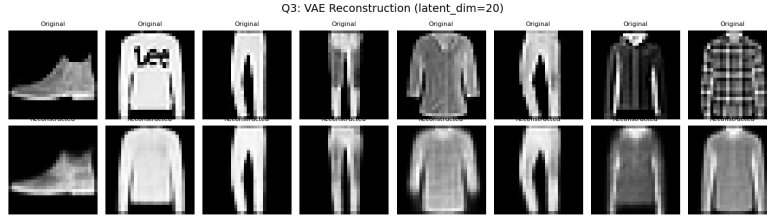


Fig. 5. Q3 VAE: Original (top) vs Reconstructed (bottom) Fashion-MNIST images.

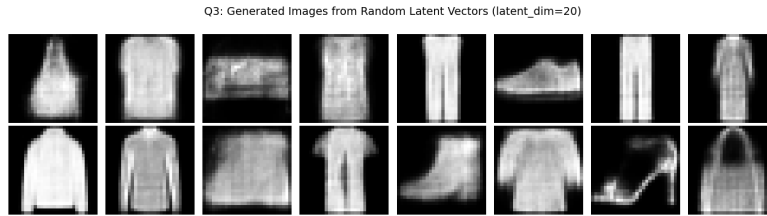


Fig. 6. Q3 VAE: Novel images generated by sampling $z \sim \mathcal{N}(0, I)$.

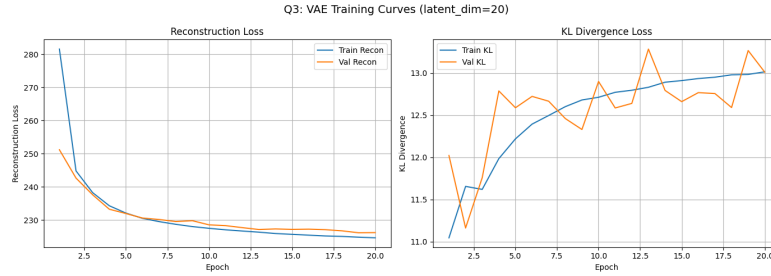


Fig. 7. Q3 VAE training curves: reconstruction loss and KL divergence.

Reconstructions preserved the global shape and category of each item (shoe, trouser, shirt) while exhibiting characteristic VAE blurriness. Novel samples drawn from the prior showed diverse, recognisable clothing silhouettes. Higher latent dimensions ($z = 50$) produced sharper reconstructions at the cost of less smooth interpolations, while $z = 2$ collapsed category structure.

4.6 Limitations

VAE outputs are blurry because the pixel-wise BCE loss penalises average error rather than perceptual quality. GANs and diffusion models address this by replacing the reconstruction term with an adversarial or score-matching objective.

5 Conclusion

This work implemented and evaluated three foundational generative architectures. The vanilla RNN encoder-decoder confirmed theoretical limitations: vanishing gradients, the context bottleneck, and the inability to generalise from small corpora produce degenerate repetitive output. These failures directly motivate attention mechanisms and transformer architectures. The convolutional DAE demonstrated robust denoising capability (PSNR up to 23.46 dB, SSIM up to 0.964), validating the encoder-bottleneck-decoder design for image restoration. The VAE confirmed the reparameterisation trick’s effectiveness for learning a regularised latent space suitable for generation. Future work would extend the translation system to LSTM with Bahdanau attention, replace the DAE’s MSE loss with perceptual loss, and compare the VAE against a GAN baseline.

References

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